Water-Use Efficiency under fertilization schemes in an elevational gradient– Preliminar results from NUMEX

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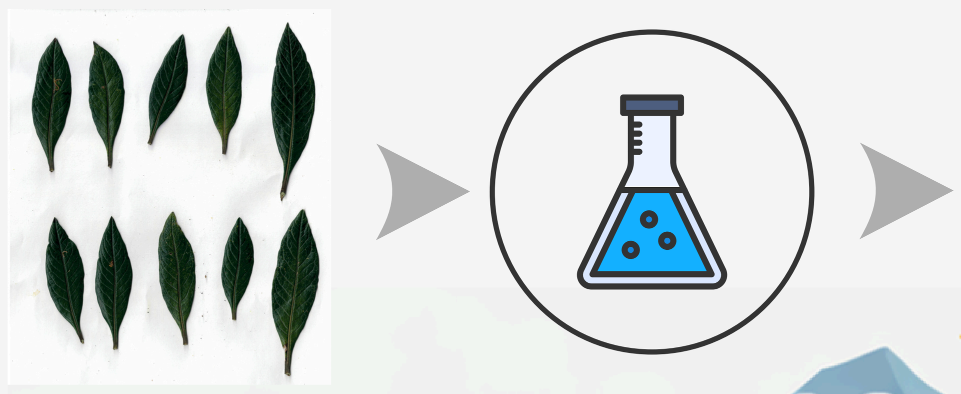
Background

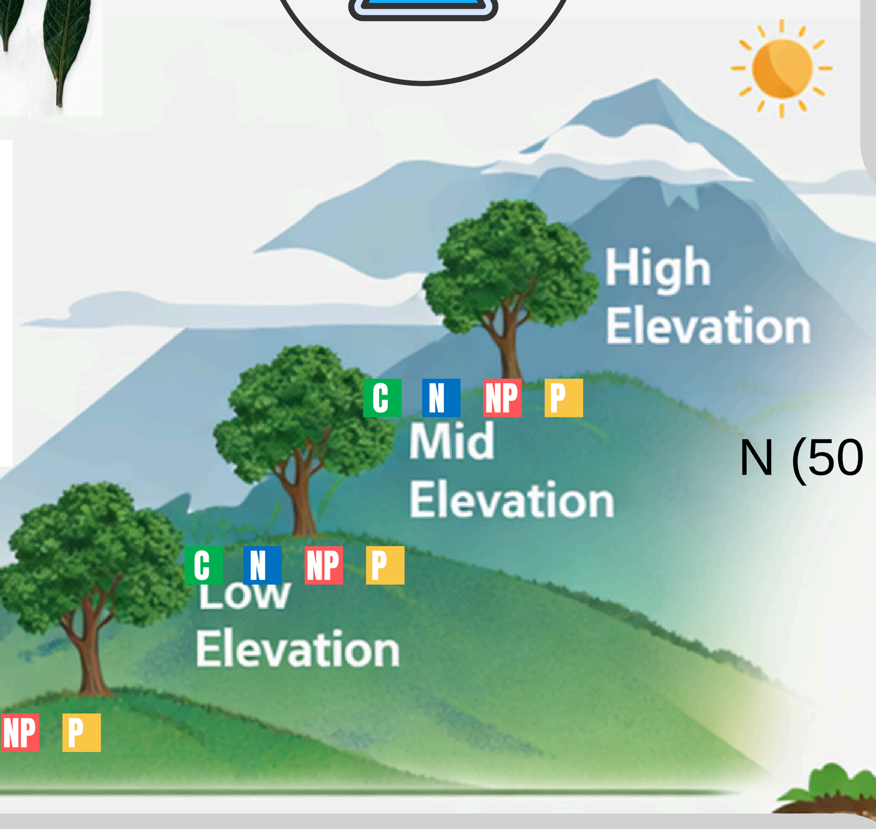
- **Intrinsic water-use efficiency (iWUE)** is a key physiological parameter that reflects how plants balance carbon assimilation with stomatal regulation of water loss.
- **NUMEX** is a long-term nutrient manipulation experiment established in 2008 in a tropical montane forest in the South of Ecuador. It was designed to evaluate ecosystem responses to sustained, moderate additions of nitrogen (N) and phosphorus (P).
- There is **limited information** on the ecophysiological responses of tropical montane forests in the Andes in terms of iWUE.

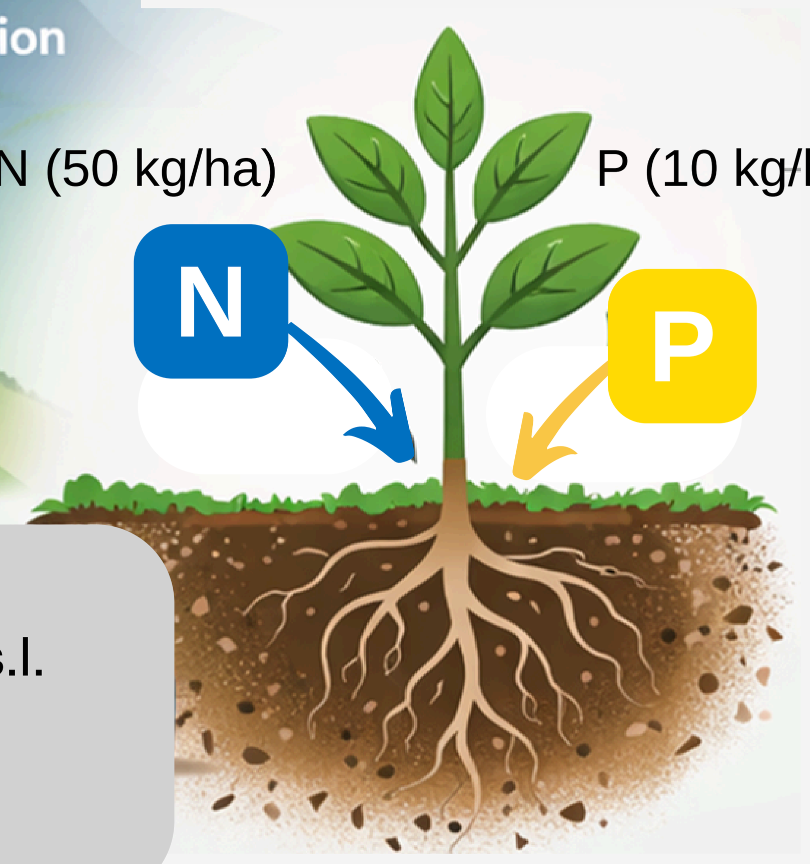
Objective

Evaluate the effect of moderate nutrient addition to **iWUE and leaf nutrients** after 18 years of fertilization across treatments in an elevational gradient.

Methodology







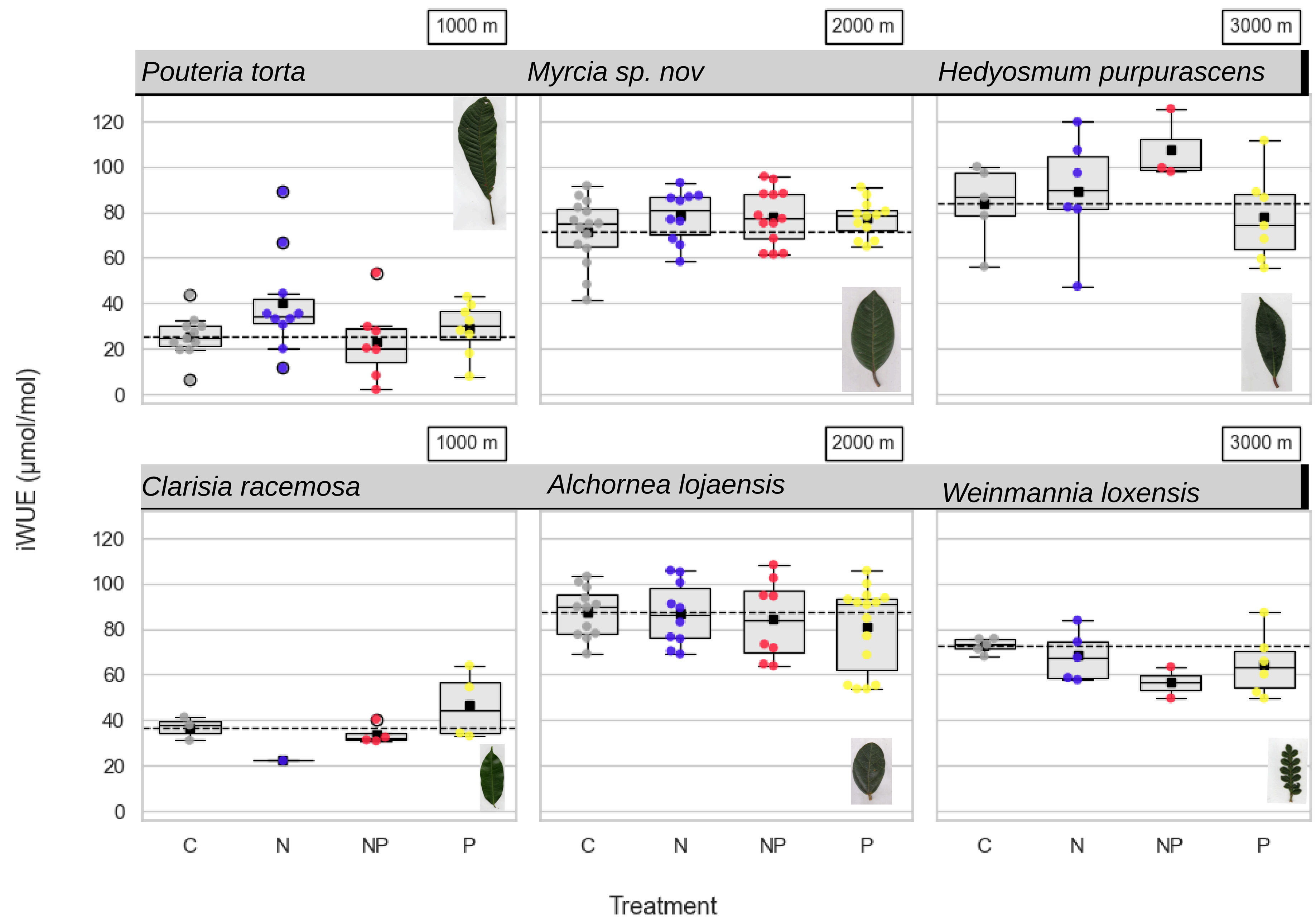
Leaf data collected:
- Specific leaf area -SLA
- Leaf nitrogen content LNC
- Leaf phosphorus content LPC
- Cellulose $\delta_{13}C$ -$\delta_{18}O$
- Leaf iWUE

- Altitude sites: 1000 m.a.s.l, 2000 m.a.s.l, and 3000 m.a.s.l.
- Full factorial design 4x4.
- Nutrient treatments : Control, N,P, N+P

Results

Response	Elevation	Species	N	N+P	P
LNC (mg/g)	1000	*Pouteria torta*			+*
		Clarisia racemosa		+*	
	2000	*Alchornea lojaensis*	+**		
		Myrcia spnov	+**	+*	
	3000	*Hedyosmum purpurascens*	+**		
		Weinmannia loxensis			
LPC (mg/g)	1000	*Pouteria torta*			
		Clarisia racemosa		+**	+**
	2000	*Alchornea lojaensis*		+***	+***
		Myrcia spnov		+***	+***
	3000	*Hedyosmum purpurascens*			+***
		Weinmannia loxensis			+***
Leaf ratio N:P	1000	*Pouteria torta*			
		Clarisia racemosa			
	2000	*Alchornea lojaensis*	+*	-***	-***
		Myrcia spnov		-***	-***
	3000	*Hedyosmum purpurascens*	+*		-*
		Weinmannia loxensis			-***

Summary of significant leaf nutrient responses to nitrogen and phosphorus addition compared to control plots in mature old-growth tropical montane forest tree species.



Conclusions

- **Nutrient additions** (N, P, and NP) **did not significantly change iWUE** compared to control plots across elevations, despite increasing leaf nutrient content in some species.
- In contrast, **iWUE increased with altitude,** suggesting that climatic factors associated with higher elevations (e.g., lower temperatures) are stronger drivers of plant water-use strategies than nutrient availability.
- The results provide **new insights** into water-use strategies of **tropical montane tree species** and highlight the importance of **long-term nutrient manipulation experiments** for addressing knowledge gaps in tropical forest eco-physiology.

Future work

Functional traits, leaf nutrients, and intrinsic water-use efficiency (iWUE) across temporal and spatial gradients in Ecuadorian montane forests.

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